

Assignment 2: A Bayesian Test of Coin Fairness

CSDS 611 — Advanced Topics and Methods of Data Science I

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Data, hypotheses, and assumptions

The observed sequence is

$$D = (H, H, H, T, T, H), \quad n = 6, \quad k = 4 \text{ heads and } 2 \text{ tails.}$$

Let θ be the probability of heads on one flip. The hypotheses in the assignment are

$$H_0 : \theta = \frac{1}{2} \quad \text{and} \quad H_A : \theta \in [0, 1] \text{ is uncertain.}$$

I assume the flips are independent Bernoulli trials conditional on a shared θ .

Under H_A , I use $\theta \mid H_A \sim \text{Beta}(1, 1)$, which is uniform on $[0, 1]$. I also set $P(H_0) = P(H_A) = 1/2$ and use the MAP rule, which chooses the hypothesis with the larger posterior probability.

Likelihood and model evidence

For the observed sequence, the likelihood is

$$p(D \mid \theta) = \theta^4(1 - \theta)^2.$$

Under H_0 ,

$$p(D \mid H_0) = \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 = \frac{1}{64}.$$

Under H_A , I average the likelihood over my $\text{Beta}(1, 1)$ prior. Its density is 1 on $[0, 1]$, so

$$\begin{aligned} p(D \mid H_A) &= \int_0^1 \theta^4(1 - \theta)^2 d\theta \\ &= \int_0^1 (\theta^4 - 2\theta^5 + \theta^6) d\theta \\ &= \frac{1}{5} - \frac{1}{3} + \frac{1}{7} = \frac{1}{105}. \end{aligned}$$

Comparison and decision

The Bayes factor for H_A against H_0 is

$$\text{BF}_{A0} = \frac{p(D \mid H_A)}{p(D \mid H_0)} = \frac{1/105}{1/64} = \frac{64}{105} \approx 0.610.$$

Thus these data favor H_0 over H_A by a factor of $105/64 \approx 1.64$. With equal prior probabilities for the hypotheses, Bayes' rule gives

$$\begin{aligned} P(H_0 \mid D) &= \frac{(1/64)(1/2)}{(1/64)(1/2) + (1/105)(1/2)} = \frac{105}{169} \approx 0.621, \\ P(H_A \mid D) &= \frac{64}{169} \approx 0.379. \end{aligned}$$

Under the choices stated above, the MAP rule selects H_0 . The evidence is modest: these six flips do not establish that the coin is fair.